Felipe Rodriguez

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EDUCATION

Georgia Institute of Technology, Atlanta, GA Expected Graduation Date: May 2028

Bachelor of Science in Mechanical Engineering – GPA: 4.00/4.00

EXPERIENCE

HyTech Racing (Formula SAE Electric) Sep. 2025 – Present

Aerodynamics & Suspension Design Member Atlanta, GA

- Built a MATLAB torque spec tool (Shigley frustum method) for the suspension-to-monocoque joints, checking joint separation, shear slip, layer yield, and bolt proof load; verified against Shigley Example 8-2 and submitted to my lead.
- Reduce CFD solve residuals by 90% by tuning body-of-influence sizing and iteration count, smoothing mesh gradients and improving flux conservation across the model.
- Support high-resolution CFD simulations in ANSYS Fluent on Georgia Tech's PACE high-performance computing cluster, configuring boundary/naming selections to optimize downforce and reduce vehicle drag.
- Fabricate structural composite aero packages using resin-infusion carbon fiber layups, precision mold tooling, debulking, sanding, and surface finishing.
- Ran a 5-case CFD sweep across rear wing angles, establishing the team's first medium-downforce CFD dataset alongside existing high-downforce data.
- Identified a wing angle that reduced drag by 28% with only a 17% downforce trade-off versus the high-downforce baseline, and presented findings to the aero lead to inform configuration selection for high-speed corners.

JNCY Jewelers Summers 2023 & 2024

3D CAD Design Intern Miami, FL

- Generated thousands of dollars in revenue by designing 12+ custom CAD models for manufacturing, working backward from lost-resin casting constraints to meet both client needs and the employer's manufacturing capabilities.
- Collaborated directly with executive leadership and clients to translate conceptual sketches into high-precision digital assemblies within tight manufacturing tolerances.
- Prepared watertight models optimized for additive manufacturing (3D resin printing) and investment casting, resolving wall thickness and shrinkage constraints.

Handshake AI Feb. 2026 – Present

AI Training Contractor – CAD & Mechanical Systems Remote

- Create and audit large datasets for training generative CAD models by filtering out subjective design choices and enforcing standard CAD conventions.
- Evaluate and debug CAD instruction sets and generative outputs for physical plausibility, geometric tolerance accuracy, and adherence to drafting conventions.

PROJECTS & LEADERSHIP

Scratch-Built RC Aircraft Apr. 2025 – Jun. 2025

Lead Aerodynamic & Structural Designer – Independent Project Miami, FL

- Derived a ~10 lbf target lift budget from airfoil-based hand calculations to size a foam-board airframe with wooden spar reinforcement around a 255g electronics payload, achieving a final aircraft weight of ~5 lbs.
- Researched and compared airfoil profiles, ultimately selecting the Clark Y airfoil for its smooth stall characteristics and stronger low-speed performance in the hover-prone flight regime typical of RC aircraft.
- Conducted flight trials confirming stable level flight at ~30% throttle before pilot error ended both attempts.

SKILLS & CERTIFICATIONS

Software & Modeling: CATIA, Rhino 3D (SubD & NURBS), SOLIDWORKS, Autodesk Fusion, Onshape, ANSYS Fluent (CFD), MATLAB, Python (Numerical Methods coursework)

Manufacturing & Prototyping: Resin-Infused Carbon Fiber Layups, Mold Tooling, Additive Manufacturing (3D Printing), Laser Cutting, Investment Casting Prep

Engineering & Core Skills: Computational Fluid Dynamics (CFD), FEA, DFM, GD&T, Tolerance Analysis, Rapid Prototyping, Technical Drawings

Certifications: Alteryx Designer Core Certified, Pre-Engineering Industry Certification (150-Hour Accredited Program)